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Effect of Different Mixed Microflora on the Performance of Thermophilic Microaerobic Pretreatment

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ABSTRACT: Recently, thermophilic microaerobic pretreatment (TMP) showed to be an efficient pretreatment method for anaerobic digestion (AD). During TMP, bacteria affiliated to phylum Firmicutes played an important role. The microbial community structure of inoculum might directly affect the performance of TMP. In this study, three different mixed microfloras [biogas slurry (BS), activated sludge (AS), and effluent from retted corn straw (ER)] were selected to explore inocula impact on the performance of TMP. Results showed that the TMP process could obviously improve the startup performance and methane yield of AD. Despite the inocula during TMP being different when the oxygen load during TMP was 5 mL/g of $VS_{substrate}$, the methane yields reached a maximum. The maximum methane yield was obtained when using ER as inoculum during TMP, which was 5.2 and 5.8% higher than those of samples using BS and AS as inocula, respectively. Accordingly, the relative abundance of phylum Firmicutes in ER was also highest, which was 77.4 and 459.5% higher than those of BS and AS, respectively. When using TMP as the pretreatment method, the inoculum selection during the TMP process is important, and the higher abundance of bacteria affiliated to phylum Firmicutes could improve the performance of TMP.

1. INTRODUCTION

Biomethane production from low-value organic waste according to anaerobic digestion (AD) has caught wide attention and is considered as an important part of the future bioeconomy.[1] Corn straw is abundant and rich in carbohydrate content, which is a potential substrate for biogas production through AD.[2] However, cellulose, hemicelluloses, and lignin of corn straw were interlinked strongly, which makes the hydrolysis process of AD inefficient.[3] Therefore, pretreatment is essential during the AD of corn straw to break the compositional and structural barriers or directly enhance the hydrolysis process.[4−7] Many pretreatment methods, including biological, mechanical, and chemical pretreatments, as well as their combinations have been proven as efficient pretreatment methods.[8−10]

Recently, microaerobic pretreatment was shown to be an attractive pretreatment method.[9,11−13] Different from other chemical or mechanical pretreatment methods, microaerobic pretreatment just needs less amount of oxygen.[14] Overall, there are two kinds of application methods to use microaeration to improve the AD efficiency: (1) in accordance with a microaerobic pretreatment step before AD and (2) direct introduction of limited oxygen into the AD digester. In comparison to other pretreatment methods, there are no additional chemical or energy inputs during microaerobic pretreatment and the application of thermophilic microaerobic pretreatment (TMP) in the biogas plant can be achieved by the addition of a TMP tank, which can directly remold the existing reactor of the biogas plant. Therefore, microaerobic pretreatment is more economic and environmentally friendly. Microaerobic pretreatment is mainly a biological process during which the activities of facultative bacteria play an important role.[15] The growth rate of facultative bacteria was higher under microaerobic conditions; therefore, more extracellular enzymes (e.g., cellulase and protease) were produced and eventually led to the higher hydrolysis process.[16,17] The higher abundance of phylum Firmicutes was reported by Lim et al.[18] as the groundwork for a higher hydrolysis process under microaerobic conditions. In our previous study, the higher abundance of bacteria affiliated to phylum Firmicutes was also demonstrated to be the groundwork for TMP.[19]

In general, the community structure of inoculum during TMP could be a crucial factor for the performance of TMP. The objective of this study was to investigate the effect of different mixed microflora on the performance of TMP. In addition, microbial community structure analyses were also given to different mixed microflora.

2. MATERIALS AND METHODS

2.1. Substrate and Inocula. The substrate used in this study was chopped corn straw, which was collected from a corn field in Pingdu (Shandong province) and sieved to a size of less than 0.5 cm. The total solids (TS) and volatile solids (VS) of corn straw are 91.87 and 89.50% (% of TS), respectively (standard methods[20] were used to test TS and VS).

In this study, there were a total of three different inocula: biogas slurry (BS), activated sludge (AS), and effluent from retted corn straw (ER). BS was collected from a 500 m^3 size of the biogas plant (Qingdao, Shandong, China), which was running at 35 °C and used corn straw as the substrate, with the TS and VS of 6.64 and 70.62% (% of TS), respectively. AS was collected from an anaerobic digester of the Tuandao Water Treatment Plant (Qingdao, Shandong, China), which was mainly used to deal with urban sewage, with the TS and VS of 5.38 and 49.60% (% of TS), respectively. ER was collected from retted corn straw after piled fermentation at ambient temperature for 1

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month, with the TS and VS of 3.35 and 55.11% (% of TS), respectively. These three inocula were selected as a result of the different substrates (corn straw or wastewater) that they dealt with or different fermentation conditions (anaerobic or aerobic) that they employed, which might give different microflora to these three inocula. All of the inocula were stored at 4 °C in a refrigerator.

2.2. TMP of Corn Straw with Different Inocula. During TMP, all of the experiments were performed in duplicate. During TMP, the total reaction volume was 1 L, among which there was 5.8 g (wet weight) of corn straw. The weight of inocula for BS, AS, and ER were 20.0, 24.7, and 39.7 g (wet weight), respectively. Each bottle was flushed with N_2 to remove oxygen and closed with a rubber stopper. Oxygen loads of 0, 5, 10, and 20 mL/g of $VS_{substrate}$ (marked with consecutive numbers 0, 1, 2, and 3) during TMP were achieved by injecting 0, 24, 48, and 96 mL of oxygen (room temperature and atmospheric pressure) with a syringe. TMP was performed in a 130 rpm shaking water bath at 55 °C. A gas chromatograph (SP 6890, Shandong Lunan, Inc., China) was used to measure the oxygen levels every 4 h during TMP. The gas chromatograph (SP 6890, Shandong Lunan, Inc., China) was equipped with a Porapak Q stainless-steel column (180 cm long and 3 mm outer diameter) and a thermal conductivity detector. The temperatures of the injector, detector, and oven were 50, 100, and 100 °C, respectively. The carrier gas was nitrogen.

2.3. Batch AD Tests. Mesophilic (37 °C) biogas productions were performed in duplicate. After TMP, the bottles were added with 20.0 g (wet weight) of BS to ensure the same inoculum during batch mesophilic AD, and the total reaction volume was 0.2 L. The biogas yield from corn straw without pretreatment was also tested (marked as WP). Batch AD experiments were also performed in a shaking water bath at 37 °C with 130 rpm. In this stage, the biogas production and composition were measured by the water replacement method and gas chromatograph (SP 6890, Shandong Lunan, Inc., China), respectively. The gas chromatograph used in this stage was the same as in section 2.2 but using argon as the carrier gas.

2.4. Model for Data Fitting. In this study, the modified Gomperz equation[21,22] was used to estimate performance parameters. The equation of the modified Gomperz equation was written as follows:

$$P(t) = P \exp\left[-\exp\left(\frac{R_m e}{P}(\lambda - t) + 1\right)\right]$$

where $P(t)$, P, R_m, λ, and t represent the cumulative methane production (mL/g of VS), methane production potential (mL/g of VS), maximum methane production rate (mL/day), lag-phase time (day), and elapsed time (day), respectively.

2.5. Microbial Community Structure Analysis. A total of 10 mL of different inocula were collected before and after the TMP process. Before microbial community structure analysis, all of the samples were stored at −80 °C in a refrigerator. Next-generation sequencing library preparations and Illumina MiSeq sequencing were conducted at GENEWIZ, Inc. (Beijing, China), which was in accordance with the reported methods.[19]

3. RESULTS AND DISCUSSION

3.1. Oxygen Consumption during TMP with Different Mixed Microflora. In our previous study, the oxygen load of 5 mL/g of $VS_{substrate}$ during TMP has been studied to be optimized when using BS as inoculum of TMP.[12] Thus, in this study, the oxygen loads during TMP were designed as 0, 5, 10, and 20 mL/g of $VS_{substrate}$ to study the effect of oxygen load during TMP with different inocula on the performance of TMP. The TMP process was performed until all of the injected oxygen was consumed, which lasted for 20 h (as shown in Figure 1). As a result of the different substrates (corn straw or wastewater), these three inocula dealt with different fermentation conditions (anaerobic or aerobic) employed and they might have different microbial structures. Therefore, the oxygen

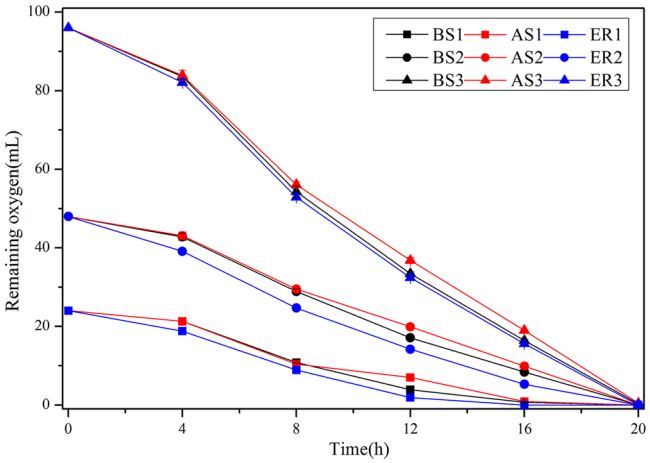


Figure 1. Oxygen consumption during TMP with different mixed microflora.

consumption rates during TMP were different. Among these three inocula, the oxygen consumption rate in ER was fastest. These could be ascribed to the aerobic fermentation condition used by ER, which led to the aerobic or facultative anaerobic bacteria being more abundant in ER, and oxygen was therefore consumed faster. To keep the same thermal (55 °C) pretreatment time, all of the TMP process lasted for 20 h.

3.2. Effect of Oxygen Loads during TMP on the Fermentation of Corn Straw with Different Mixed Microflora. Oxygen introduced during TMP was mainly used to support the growth of facultative bacteria, which, in turn, produced more extracellular enzymes (e.g., cellulase and protease).[16] The produced extracellular enzymes acted on corn straw. Finally, more available sugars were yielded, and the structure of corn straw was partly destroyed, which would be beneficial for the following AD process. However, oxygen introduced during TMP would also consume some of the substrate. A balance between facultative bacteria growth and substrate consumption should be found. The methane-producing characteristics of corn straw after TMP with different inocula are shown in Figure 2. Despite the inocula during TMP being different, the relationship between oxygen load during TMP and cumulative methane yield showed the same trend (as shown in Figure 3). The maximum methane yield was consistently obtained at the oxygen load of 5 mL/g of $VS_{substrate}$ during TMP. A further increase of the oxygen load during TMP might have a negative effect on the methane yield, which may be due to substrate consumption by facultative organisms[9] under high oxygen levels. However, methane yields of thermophilic microaerobic pretreated groups were always higher than that of the group without pretreatment. Microaerobic pretreatment of cellulosic substrates was less reported by other authors. Among the limited studies, uneven results were obtained. Mshandete et al.[13] obtained a 26% higher methane yield from sisal pulp after 9 h of microaerobic pretreatment. However, according to Diaz et al.,[23] the microaerobic condition even had no effect on the methane yield and hydrolytic activity, except shortening the lag phase during the AD of cellulose. During the two-phase solid−liquid AD of vegetable and flower waste, Zhu et al.[17] reported that a sufficient oxygen supply could just enhance the hydrolysis of an easily biodegradable substrate and an insufficient oxygen supply would lead to instability of the reactor and a decreased AD performance.

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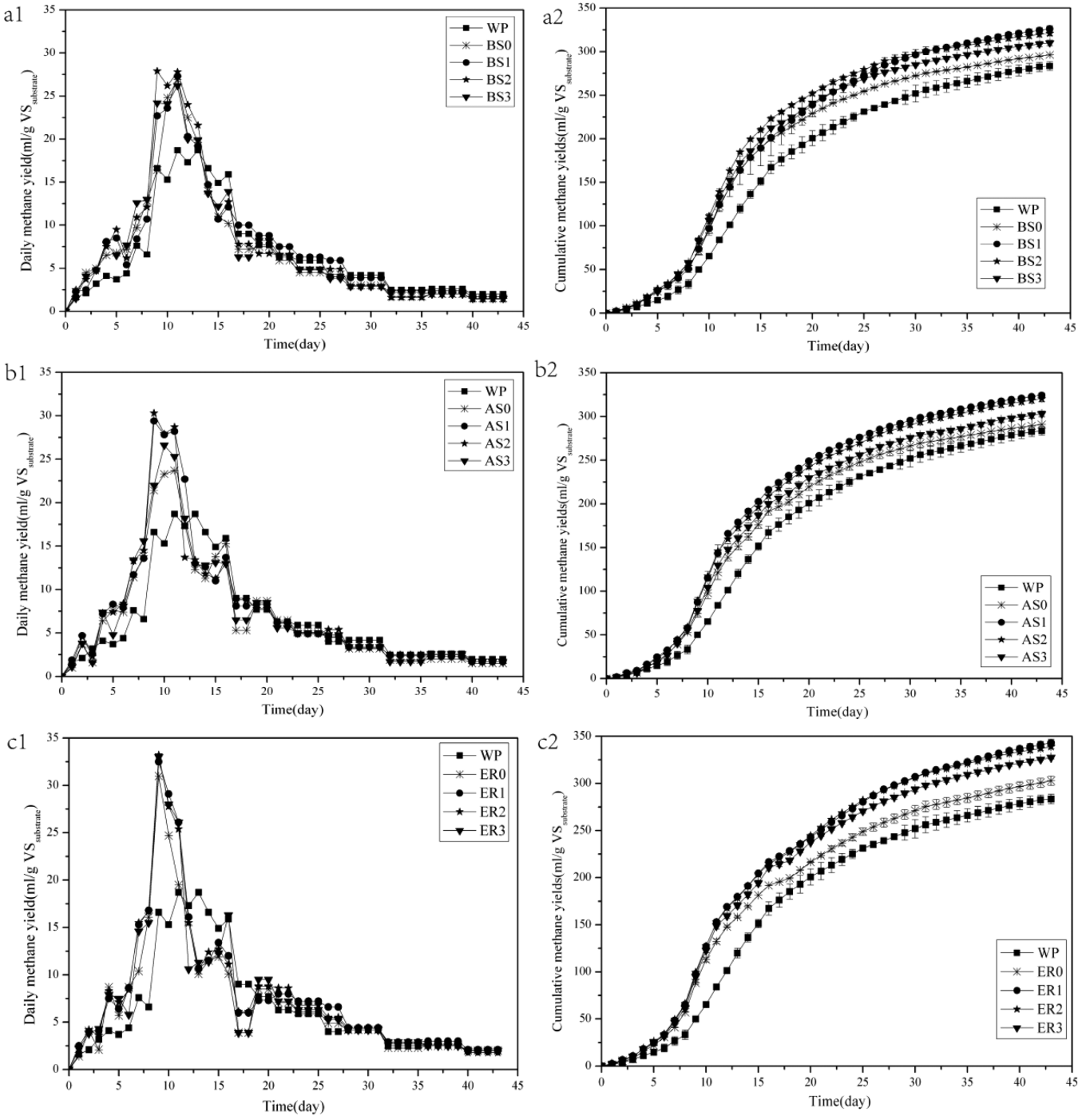


Figure 2. Methane-producing characteristics from corn straw with different inocula during TMP (a1, b1, and c1, daily methane yield from corn straw with BS, AS, and ER as inocula during TMP, respectively; a2, b2, and c2, cumulative methane yield from corn straw with BS, AS, and ER as inocula during TMP, respectively).

3.3. Effect of Different Mixed Microflora during TMP on the Performance of TMP. Because the maximum cumulative methane yield was consistently obtained at the oxygen load of 5 mL/g of $VS_{substrate}$ during TMP, the methane-producing data at the oxygen load of 5 mL/g of $VS_{substrate}$ during TMP (BS1, AS1, and ER1) were selected to study the effect of inocula during TMP on the performance of TMP.

3.3.1. Methane-Producing Characteristic of Thermophilic Microaerobic Pretreated Corn Straw with Different Mixed Microflora during TMP and Modified Gomperz Equation Fitting. The methane-producing data of thermophilic microaerobic pretreated corn straw with different mixed microflora during TMP are shown in Figure 2. The daily methane yield of thermophilic microaerobic pretreated corn straw increased sharply at the initial stage of AD. At the 9th day of AD, daily methane yields of AS1 and ER1 reached maxima, which were 30.1 and 32.5 mL/g of $VS_{substrate}$ day^{-1} (37 °C and normal pressure), respectively. By comparison, the maximum daily methane yield of BS1 was obtained at the 11th day of AD, which was only 27.3 mL/g of $VS_{substrate}$ day^{-1} (37 °C and normal pressure). In comparison to WP, the methane yield of BS1, AS1, and ER1 were more centralized and rapid. The

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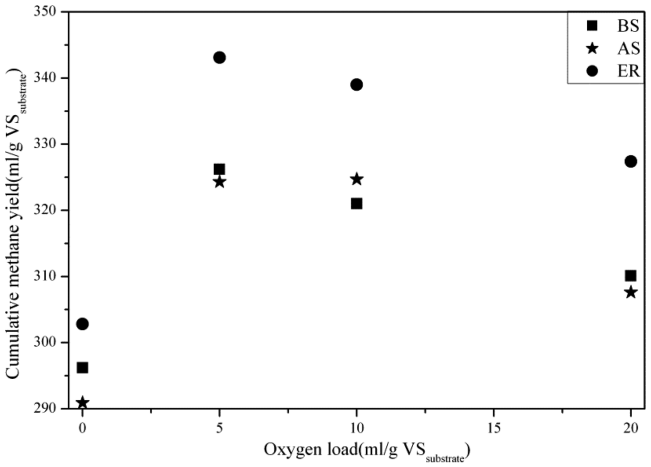


Figure 3. Relationship between oxygen loads during TMP and cumulative methane yields.

average methane production rates of BS1, AS1, ER1, and WP were 7.6, 7.5, 8.0, and 6.6 mL/g of $VS_{substrate}$ day^{-1} (37 °C and normal pressure), which means that the methane production efficiencies could obviously be improved by the TMP process before AD. In addition, using ER as inoculum of TMP obtained the maximum methane production rate.

In this study, statistical data analysis of the cumulative methane yield was carried out with the software of Microsoft Office Excel 2007. A pairwise comparison of the cumulative methane yield was conducted by the *t* test. The level of significance was set to at least 0.05. Results showed that cumulative methane yields from thermophilic microaerobic pretreated corn straw were significant higher ($p < 0.05$) than that of corn straw without pretreatment. In addition, the cumulative methane yields between different groups also had a significant difference ($p < 0.05$). The maximum cumulative methane yield of 343.1 mL/g of $VS_{substrate}$ (37 °C and normal pressure) was obtained from ER1, which was 5.2, 5.8, and 21.0% higher than those of BS1, AS1, and WP, respectively.

In this study, the modified Gomperz equation was used to analyze the fermentation kinetics of corn straw. The parameters of the modified Gomperz equation fitting experimental data are shown in Table 1. Time elapsed until a significant production of methane found in the batch assay was interpreted as the lag-phase time (λ). A higher λ means a slow startup. All λ of thermophilic microaerobic pretreated samples were shorter than that of WP, which means that the corn straw after TMP has a much higher startup. In addition, the minimum lag-phase time was obtained from ER1, which was almost 1.947 days shorter than that of WP. The same trend in lag-phase time was also reported by Diaz et al.,[23] who demonstrated a shorter lag-phase time in studying the effect of microaerobic conditions on the degradation kinetics of cellulose. The total methane production potential (P) of thermophilic microaerobic pre-treated samples was higher than WP, which was quite agreeable with experimental results. The maximum methane production potential was also obtained from ER1, which was 20.3% higher than WP. In terms of lag-phase time and total methane production potential, the ER was also the ideal inoculum of TMP among these three inocula.

Table 1. Parameters of the Modified Gomperz Equation Fitting Experimental Data

group	P (mL/g of $VS_{substrate}$)	λ (day^{-1})	R^2
WP	277 ± 2	5.349 ± 0.004	0.997
BS1	320 ± 2	4.295 ± 0.004	0.996
AS1	313 ± 3	4.093 ± 0.006	0.995
ER1	333 ± 4	3.402 ± 0.006	0.991

3.3.2. VS Removal Efficiencies of Thermophilic Microaerobic Pretreated Corn Straw with Different Mixed Microflora during TMP. The biogas produced during AD was initially from the VS of the substrate, and the removal efficiency of VS represented the degradation extent of the substrate,[24] such that the efficiency of AD could be evaluated in terms of VS removal efficiency.[25] In this study, the VS removal efficiency of inocula was assumed to be the same in the blank (only inocula during AD) and experimental group (inocula and substrate); thus, the VS removal efficiency was calculated as follows: VS removal efficiency = [initial VS of the substrate − (total final VS of inocula and the substrate − final VS of inocula)]/(initial VS of the substrate) × 100%.

The VS removal efficiencies of BS1, AS1, ER1, and WP were 66.1 ± 1.2, 65.9 ± 0.7, 72.0 ± 0.3, and 62.4 ± 0.6%, respectively. VS removal efficiency data were quite agreeable with the methane production data, and the maximum VS removal efficiency was also obtained from ER1, which was 8.9, 9.3, and 15.4% higher than those of BS1, AS1, and WP, respectively. In terms of VS removal efficiency, the ER was the ideal inoculum of the TMP process among these three inocula.

3.4. Microbial Community Structure Analysis. The microbial community structures of inocula before and after the TMP process were analyzed, which are shown in Figure 4. Before TMP, phylum Bacteroidetes was dominant in both BS and ER, which account for 51.5 and 43.69% of the total sequence, respectively. Relatively, phyla Chloroflexi and Proteobacteria were dominant in AS, which account for 26.41 and 26.4% of the total sequence, respectively. The relative abundance of phylum Firmicutes in ER was 17.01%, which was the most among these three inocula and 77.4 and 459.5% higher than those of BS and AS, respectively. After the TMP process, phylum Firmicutes became dominant in all three inocula. The relative abundances of phylum Firmicutes in ER, BS, and AS were 75.11, 73.15, and 62.2%, respectively, which demonstrated the rapid growth and enrichment of bacteria affiliated to phylum Firmicutes during the TMP process. Bacteria affiliated to phylum Firmicutes were reported to produce extracellular enzymes and play an important role during microaerobic pretreatment.[18,19,26] The higher relative abundance of phylum Firmicutes means the higher ability of ER to produce extracellular enzymes. In addition, a large percentage of bacteria affiliated to phylum Firmicutes is facultative or aerobic bacteria, which would have a quick growth rate under microaerobic conditions. The higher relative abundance of phylum Firmicutes of ER also means that facultative or aerobic bacteria had a shorter doubling time, which would also produce more extracellular enzymes. Thus, more cellulosic substrates could be hydrolyzed to a simple substrate during the TMP process. More compositional and structural barriers were destroyed in corn straw, and therefore, a higher methane production rate was obtained. Higher bacteria affiliated to the phylum Firmicutes under microaerobic conditions were also reported by Lim et al.,[18] which was also supposed to be the groundwork for the higher hydrolysis under microaerobic conditions. Similarly, the relative abundance of

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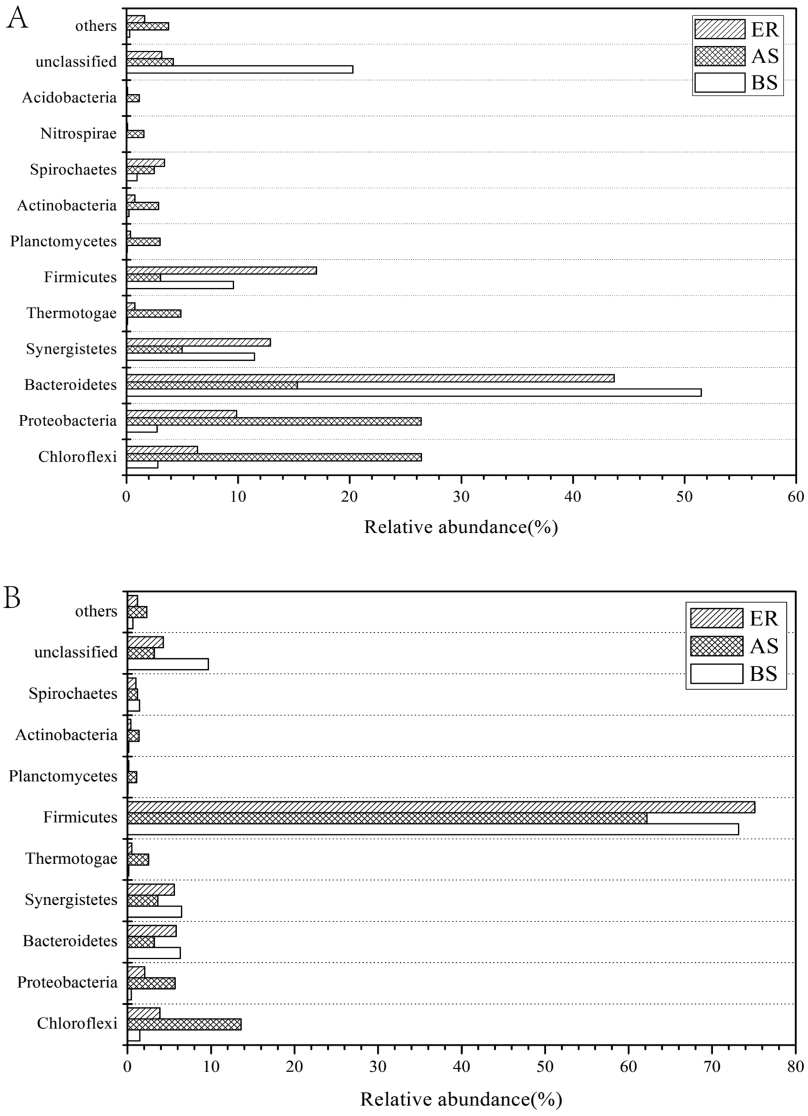


Figure 4. Microbial community structure of inocula at the phylum level. Phyla making up less than 1% of the total sequence in each sample were classified as others. The microbial community structure data of BS was cited from ref 1. (A) Microbial community structure of inocula before TMP and (B) microbial community structure of inocula after TMP.

phylum Firmicutes was also detected to be higher under microaerobic conditions in our previous study.[27]

4. CONCLUSION

The methane yield and VS removal efficiency of corn straw could obviously be improved by TMP. In addition, the startup performance of AD was also obviously improved. Among the three inocula, ER was shown to be the optimized inoculum during the TMP process, which could be attributed to the higher relative abundance of phylum Firmicutes in ER. TMP is an efficient pretreatment method during the AD of a cellulosic substrate, such as corn straw, and the microbial community of inoculum during TMP plays an important role. Inoculum with a higher abundance of phylum Firmicutes could be favorable for better performance of TMP.

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Notes

The authors declare no competing financial interest.

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